

Solutions to Homework #5

Problem 1

First note that

$$x_1[n] = \left(\frac{1}{3}\right)^n u[n].$$

We have shown that

$$\delta[\langle n \rangle_N] \xleftrightarrow{\text{DTFT}} \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right).$$

Let $S(e^{j\omega})$ denote the sampled version of $X_1(e^{j\omega})$ given by

$$S(e^{j\omega}) = X_1(e^{j\omega}) \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right).$$

Then the inverse DTFT of $S(e^{j\omega})$ is written as

$$\begin{aligned} s[n] &= \frac{1}{2\pi} \int_{[0, 2\pi)} S(e^{j\omega}) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{[0, 2\pi)} X_1(e^{j\omega}) \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{[0, 2\pi)} X_1(e^{j\omega}) \frac{2\pi}{N} \sum_{k=0}^{N-1} \delta\left(\omega - \frac{2\pi k}{N}\right) e^{j\omega n} d\omega \\ &= \frac{1}{N} \sum_{k=0}^{N-1} \int_{[0, 2\pi)} X_1(e^{j\omega}) \delta\left(\omega - \frac{2\pi k}{N}\right) e^{j\omega n} d\omega \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X_2[k] e^{j\frac{2\pi}{N}kn}, \end{aligned}$$

which coincides with the inverse DFT of $X_2[k]$ for $n = 0, 1, \dots, N-1$. Furthermore, by the convolution property of DTFT, the inverse DTFT of $S(e^{j\omega})$ is expressed as

$$s[n] = x_1[n] * \delta[\langle n \rangle_N].$$

Since $x_1[n] = 0$ for $n < 0$, we have

$$s[n] = \sum_{l=0}^{\infty} x_1[n + lN] = \sum_{l=0}^{\infty} \left(\frac{1}{3}\right)^{n+lN} = \frac{1}{3^n} \sum_{l=0}^{\infty} \left(\frac{1}{3^N}\right)^l = \frac{3^{-n}}{1 - 3^{-N}}.$$

for $n = 0, 1, \dots, N-1$, and $x_2[n] = 0$ for all other n .

Problem 2

Recall that the N -point DFT of the N -point sequence $x[n]$ is given by

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad k = 0, 1, \dots, N-1,$$

where $W_N^k = e^{-j\frac{2\pi}{N}k}$. In particular, for $k = 0$, we have

$$X[0] = \sum_{n=0}^{N-1} x[n].$$

- (a) Suppose that $x[n] = -x[N-1-n]$. For even N , all elements of $x[n]$ will cancel with an antisymmetric component. For odd N , all elements have a counterpart with opposite sign. However, $x[(N-1)/2]$ must also be zero. Therefore, $x[n] = -x[N-1-n]$ implies $X[0] = 0$.
- (b) Suppose that $x[n] = x[N-1-n]$ and N is even. Then

$$\begin{aligned} X[N/2] &= \sum_{n=0}^{N-1} x[n] W_N^{(N/2)n} \\ &= \sum_{n=0}^{N-1} x[n] (-1)^n \\ &= x[0] - x[1] + x[2] - x[3] + \dots + x[N-2] - x[N-1]. \end{aligned}$$

The symmetry condition $x[n] = x[N-1-n]$ implies

$$\begin{aligned} x[0] &= x[N-1] \\ x[1] &= x[N-2] \\ &\vdots \end{aligned}$$

Therefore, $X[N/2] = 0$.

Problem 3

- (a) Recall $h[n] = \delta[n] - \frac{1}{2}\delta[n-n_0]$. Then

$$H(z) = 1 - \frac{1}{2}z^{-n_0}$$

and

$$H[k] = \sum_{n=0}^{N-1} \left(\delta[n] - \frac{1}{2}\delta[n-n_0] \right) e^{-j\frac{2\pi kn}{N}} = 1 - \frac{1}{2}e^{-j\frac{2\pi kn_0}{N}} = 1 - \frac{1}{2}e^{-j\frac{\pi k}{2}}.$$

Here we used $N = 4n_0$.

(b) The inverse system is given by

$$H_i(z) = \frac{1}{H(z)} = \frac{1}{1 - \frac{1}{2}z^{-n_0}} = F(z^{n_0}),$$

where

$$F(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}.$$

We know that the inverse z -transform of $F(z)$ is given by

$$f[n] = 2^{-n}u[n].$$

Since $h_i[n]$ is the upsampled version of $f[n]$, we have

$$h_i[n] = \begin{cases} 2^{-n/n_0}, & \frac{n}{n_0} \in \mathbb{Z}, n \geq 0, \\ 0, & \text{else} \end{cases}$$

This is an IIR filter.

(c) Note that

$$G[k] = \frac{1}{H[k]} = H_i(e^{j\omega})|_{\omega=\frac{2\pi k}{N}}.$$

Let $\tilde{g}[n]$ denote the periodization of $g[n]$, that is $\tilde{g}[n] = g[n]$ for $n = 0, 1, \dots, N-1$ and $\tilde{h}[n+N] = \tilde{g}[n]$. Then, for $n = 0, 1, \dots, N-1$, $\tilde{g}[n]$ is computed by

$$\begin{aligned} g[n] &= \frac{1}{N} \sum_{k=0}^{N-1} G[k] e^{j2\pi kn/N} \\ &= \frac{1}{2\pi} \int_{-\epsilon}^{2\pi-\epsilon} \frac{2\pi}{N} \sum_{k=0}^{N-1} \delta\left(\omega - \frac{2\pi k}{N}\right) H_i(e^{j\omega}) e^{j\omega kn} d\omega \\ &= \frac{1}{2\pi} \int_{-\epsilon}^{2\pi-\epsilon} \frac{2\pi}{N} \sum_{m=-\infty}^{\infty} \sum_{k=0}^{N-1} \delta\left(\omega - 2\pi m - \frac{2\pi k}{N}\right) H_i(e^{j\omega}) e^{j\omega kn} d\omega \\ &= \mathcal{F}_{\text{DTFT}}^{-1} \left\{ \frac{2\pi}{N} \sum_{m=-\infty}^{\infty} \sum_{k=0}^{N-1} \delta\left(\omega - 2\pi m - \frac{2\pi k}{N}\right) H_i(e^{j\omega}) \right\}, \end{aligned}$$

where $\epsilon > 0$ is a small constant. Here the summation over m makes the signal periodic with 2π so that it becomes a valid DTFT. Recall that

$$\mathcal{F}_{\text{DTFT}} \left\{ \sum_{l=-\infty}^{\infty} \delta[n - lN] \right\} = \frac{2\pi}{N} \sum_{m=-\infty}^{\infty} \sum_{k=0}^{N-1} \delta\left(\omega - 2\pi m - \frac{2\pi k}{N}\right).$$

Therefore, in the time domain, we have

$$\tilde{g}[n] = h_i[n] * \sum_{l=-\infty}^{\infty} \delta[n - lN] = \sum_{l=-\infty}^{\infty} h_i[n - lN].$$

We only consider $\tilde{g}[n]$ for $n = 0, 1, \dots, N-1$. Since $h_i[n]$ is causal, for $n = 0, 1, \dots, N-1$,

$$\begin{aligned}
g[n] &= \tilde{g}[n] \\
&= \sum_{l=0}^{\infty} h_i[n + 4n_0 l] \\
&= \begin{cases} \sum_{l=0}^{\infty} 2^{-n/n_0} 2^{-4l}, & n/n_0 \in \mathbb{Z}, 0 \leq n \leq N-1, \\ 0, & \text{else} \end{cases} \\
&= \begin{cases} \frac{2^{-n/n_0}}{1-2^{-4}}, & n/n_0 \in \mathbb{Z}, 0 \leq n \leq N-1, \\ 0, & \text{else.} \end{cases}
\end{aligned}$$

Alternatively,

$$g[n] = \frac{1}{1-2^{-4}} \cdot h_i[n](u[n] - u[n-4n_0]).$$

This means that the approximate FIR inverse filter is obtained from the actual inverse filter $h_i[n]$ by multiplying $\frac{1}{1-2^{-4}}$ and truncating it up to the first $4n_0$ samples.

(d) Finally, we compute the convolution of $g[n]$ and $h[n]$, which is

$$g[n] * h[n] = g[n] - \frac{1}{2}g[n-n_0].$$

Note that

$$\frac{1}{2}g[n-n_0] = \begin{cases} \frac{2^{-n/n_0}}{1-2^{-4}}, & n/n_0 \in \mathbb{Z}, n_0 \leq n \leq 5n_0-1, \\ 0, & \text{else.} \end{cases}$$

Therefore,

$$g[n] * h[n] = g[n] - \frac{1}{2}g[n-n_0] = \frac{1}{1-2^{-4}}\delta[n] - \frac{2^{-5}}{1-2^{-4}}\delta[n-5n_0].$$

In general, if $N = Kn_0$, we have

$$g[n] * h[n] = \frac{1}{1-2^{-K}}\delta[n] - \frac{2^{-(K+1)}}{1-2^{-K}}\delta[n-(K+1)n_0].$$

Therefore, as we increase K , $g[n] * h[n]$ approaches to $h_i[n] * h[n] = \delta[n]$.

Problem 4

Since $x[n]$ is supported on $\{0, 1, \dots, N-1\}$, the upsampled signal $v[n]$ is supported on $\{0, 1, \dots, 4N-1\}$. Consequently, sampling the DTFT of $v[n]$ on a uniform grid of size $4N$ does not introduce any time aliasing.

After upsampling, we wish to retain only the non-aliased spectral components in the ranges

$$\{0, 1, \dots, \frac{N}{2} - 1\} \quad \text{and} \quad \{4N - \frac{N}{2}, \dots, 4N - 2, 4N - 1\}.$$

Therefore, we set the magnitude response to zero outside these frequency indices:

$$|H[k]| = \begin{cases} 1 & k \in \{0, 1, \dots, \frac{N}{2} - 1\} \cup \{4N - \frac{N}{2}, \dots, 4N - 2, 4N - 1\} \\ 0 & \text{otherwise.} \end{cases}$$

We choose the phase response to satisfy

$$\angle H[k] = \frac{\pi k}{4}$$

so that

$$H[k] = \begin{cases} e^{j\frac{\pi k}{4}} & k \in \{0, 1, \dots, \frac{N}{2} - 1\} \cup \{4N - \frac{N}{2}, \dots, 4N - 2, 4N - 1\} \\ 0 & \text{otherwise.} \end{cases}$$

This construction ensures that the impulse response $h[n]$ is supported on $\{0, 1, \dots, N-1\}$ and is symmetric about its midpoint, yielding a generalized linear-phase filter.